Non-Local Magic Resources for Fermionic Gaussian States

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Non-local magic quantifies the nonstabilizerness that local unitaries cannot remove, but its computation requires an intractable optimization. For fermionic Gaussian states and contiguous bipartitions, we prove a closed formula for the Rényi-2 non-local stabilizer entropy over local Gaussian unitaries, set by the reduced Majorana covariance matrix. We derive a Page-like curve for random states, logarithmic scaling at the XY critical point, and a quasiparticle picture after quenches. Our result enables measuring fermionic Gaussian-restricted non-local magic via fermionic shadow tomography.

Introduction.—Entanglement and magic, or nonstabilizerness, are two fundamental notions of quantum complexity that capture complementary aspects of what makes a many-body system genuinely quantum [1–3]. States with low entanglement admit efficient classical simulation via tensor-network methods [4], whereas states with low magic are accessible to stabilizer-based techniques such as tableau simulation [5, 6] or Pauli propagation [7, 8]. While entanglement is a well-characterized concept in many-body physics [9, 10], the role of magic has only recently come into focus, driven by the development of stabilizer entropies [11, 12], scalable probes of nonstabilizerness whose computational cost is comparable to that of entanglement entropy [13–15] and which have an important operational meaning [16–18]. This advance has stimulated a broad program of research, from equilibrium phenomena [19–23] to quantum-circuit and Hamiltonian dynamics [24–32].

A central open question concerns the interplay between entanglement and magic [33–35]. The recently introduced *non-local magic* [36–39] addresses precisely this problem: it isolates the irreducible nonstabilizerness that cannot be removed by local basis changes, thereby separating genuinely correlated magic from purely local contributions. However, non-local magic is defined through a minimization over all local unitaries acting on the full Hilbert space, rendering it intractable beyond a few qubits. Although estimation schemes exist for matrix-product states [36], where entanglement remains bounded, which broader classes of quantum matter admit an analytic characterization of non-local magic, especially in regimes with extensive entanglement, has remained an open question.

Here, we answer this question for fermionic Gaussian states and unitaries. We introduce the fermionic Gaussian-restricted non-local (GNL) magic. We prove (cf. Theorem 1) that the GNL magic of order $\alpha = 2$ admits a closed-form expression (Eq. (5)) in terms of the eigenvalues of the reduced Majorana covariance matrix on a contiguous subsystem of size ℓ, computable with $\mathrm{poly}(\ell)$ resources. The key insight behind this result is that the minimization over local *Gaussian* unitaries is solved exactly by the BCS canonical form [40]. This is in stark contrast with the total magic, which requires Majorana Monte Carlo [41] or perfect sampling [42], both converging in general with an exponential number of samples. We leverage this result to obtain a series of rigorous physical insights: an exact Page-like curve for typical fermionic Gaussian states, corroborated numerically by SYK_2 eigenstates [41, 43–46]; logarithmic scaling of non-local magic at the XY critical point, captured analytically via the Fisher–Hartwig approach; a quasiparticle picture [47] for magic spreading after quantum quenches; and diffusive growth in brickwork Gaussian random circuits. Because our result relies solely on two-point correlation functions, it also provides a scalable route for the experimental estimation of Gaussian-restricted non-local magic in large-scale quantum processors via fermionic shadow tomography [48] (End Matter).

Methods.—We refer to Refs. [49–52] for pedagogical introductions. We consider fermionic Gaussian systems on N qubits in a lattice $\Lambda = \{1, \dots, N\}$, described by a Hilbert space of dimension $d = 2^N$. The natural degrees of freedom are the $2N$ Majorana operators γ_j, obtained via the Jordan–Wigner transformation $\gamma_{2j-1} = \big(\prod_{m=1}^{j-1} Z_m\big) X_j$ and $\gamma_{2j} = \big(\prod_{m=1}^{j-1} Z_m\big) Y_j$. Fermionic Gaussian unitaries (FGU), or matchgates, are the operators that act adjointly on Majorana as rotations $U_O \gamma_j U_O^\dagger = \sum_m O_{mj} \gamma_m$, with $O \in \mathrm{O}(2N)$; up to the reflection implemented by $X_1 = \gamma_1$, they are generated by the fermionic Gaussian Hamiltonians

$$H = \frac{i}{4} \sum_{m,n=1}^{2N} h_{m,n} \gamma_m \gamma_n \,, \tag{1}$$

where $h = -h^T$ is a $2N \times 2N$ antisymmetric matrix. A pure fermionic Gaussian state (FGS), or free fermionic state, is defined by $|\Psi_O\rangle = U_O |\mathbf{0}\rangle$ for the fermionic

Gaussian unitary induced by $O \in \mathrm{O}(2N)$, and $|\mathbf{0}\rangle\langle\mathbf{0}| = d^{-1}\prod_{m=1}^{N}(\mathbb{I}-i\gamma_{2m-1}\gamma_{2m})$ is the fermionic vacuum state. Denoting for any pure state $|\Psi\rangle$ the covariance matrix $\Gamma_{mn} = i\langle\Psi|[\gamma_m,\gamma_n]|\Psi\rangle/2$, fermionic Gaussian states are fully characterized by the covariance matrix: $|\Psi_O\rangle\langle\Psi_O|$ is uniquely fixed by $\Gamma = O\Gamma_0 O^T$, with $\Gamma_0 = \bigoplus_{m=1}^{N}\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ the vacuum covariance matrix. For pure fermionic Gaussian states and the bipartition $\Lambda = A \sqcup B$ with $|A| = \ell \leq N/2$ (made precise below), Ref. [40] showed that local Gaussian transformations $O_A \oplus O_B$ of the Majorana modes are sufficient to bring the state into a canonical modewise form of independent two-mode BCS-like pairs

$$|\Psi_{\mathrm{can}}\rangle := \prod_{i=1}^{\ell}\left(\cos\theta_i + \sin\theta_i\, c_i^\dagger c_{\ell+i}^\dagger\right)|\mathbf{0}\rangle\ , \tag{2}$$

where $c_j = (\gamma_{2j-1} + i\gamma_{2j})/2$ annihilates the vacuum, $c_j|\mathbf{0}\rangle = 0$, and $\cos 2\theta_i = \lambda_i$ are the eigenvalues introduced below.

Non-local magic and its fermionic counterpart.—We recall that the stabilizer entropy of order α is defined for a pure state $|\Psi\rangle$ as $M_\alpha(|\Psi\rangle) = (1-\alpha)^{-1}\log_2\zeta_\alpha(|\Psi\rangle)$ [11], where $\zeta_\alpha = d^{-1}\sum_{P\in\mathcal{P}_N}\langle\Psi|P|\Psi\rangle^{2\alpha}$ is the stabilizer purity and $\mathcal{P}_N = \{\mathbb{I},X,Y,Z\}^{\otimes N}$ is the set of N-qubit Pauli strings. This quantity is a magic monotone for integer $\alpha \geq 2$ [12, 53]. Throughout this work, we consider the bipartition $\Lambda = A \sqcup B$ of the chain into the contiguous blocks $A = \{1,\dots,\ell\}$ and $B = \{\ell+1,\dots,N\}$. The non-local magic (NLM) of a pure state is defined as

$$M_\alpha^{\mathrm{NL}} := \min_{U_A,U_B} M_\alpha\left[(U_A\otimes U_B)|\Psi\rangle\right]\ , \tag{3}$$

where the minimization is over $\mathrm{U}(d_A)\otimes\mathrm{U}(d_B)$ with $d_A = 2^\ell$ and $d_B = 2^{N-\ell}$. This is the magic that cannot be erased or extracted by local unitary operations. This notion is cognate to (but distinct from) that of *long-range magic*, that is, the non-stabilizerness that cannot be removed by finite-depth local unitary circuits [54]. In Ref. [36], it was shown that for $\alpha = 2$ a reliable proxy for non-local magic can be obtained from the Schmidt spectrum of the state over the same bipartition. To study fermionic Gaussian states and their non-local magic, we introduce the fermionic Gaussian-restricted non-local magic

$$M_\alpha^{\mathrm{GNL}} := \min_{U_{O_A},U_{O_B}} M_\alpha\left[(U_{O_A}\otimes U_{O_B})|\Psi\rangle\right], \tag{4}$$

where now the minimization is over the Gaussian unitaries induced by $\mathrm{O}(2\ell)\oplus\mathrm{O}(2(N-\ell))$, hence $M_\alpha^{\mathrm{NL}} \leq M_\alpha^{\mathrm{GNL}}$. Since A and B are contiguous blocks of the Jordan–Wigner chain, $U_{O_A}\otimes U_{O_B}$ is itself a Gaussian unitary of the whole chain, induced by $\mathrm{O}(2\ell)\oplus\mathrm{O}(2(N-\ell))$ [Supplemental Material (SM) [55], Sec. S1]. Throughout this work our main interest is the case $\alpha = 2$, the smallest Rényi index for which the stabilizer entropy is a magic monotone [12, 53], and the one we implicitly refer to when writing simply *Gaussian-restricted non-local magic*.

Results.—Let $\{\lambda_i\}_{i=1,\dots,\ell}$ denote the non-negative eigenvalues of the $2\ell\times 2\ell$ reduced covariance matrix $i\Gamma_A := i\Gamma|_{\mathcal{I}_A}$ of the Majorana modes restricted to A, $\mathcal{I}_A := \{2m-1, 2m \mid m\in A\}$.

Theorem 1 (Gaussian-restricted non-local magic of Gaussian states at $\alpha = 2$)**.** *Let $|\Psi_O\rangle$ be an arbitrary pure fermionic Gaussian state on N modes and let $\Lambda = A\sqcup B$ be the bipartition into the contiguous blocks $A = \{1,\dots,\ell\}$ and $B = \{\ell+1,\dots,N\}$. Then*

$$\begin{aligned} M_2^{\mathrm{GNL}}(|\Psi_O\rangle) &= \sum_{i=1}^{\ell}\mathfrak{m}_2(\lambda_i^2)\ ,\\ \mathfrak{m}_2(x) &= -\log_2\left(1-x+x^2\right), \end{aligned} \tag{5}$$

or, equivalently, in terms of the stabilizer purity,

$$\begin{aligned} \max_{\substack{O_A\in\mathrm{O}(2\ell)\\ O_B\in\mathrm{O}(2(N-\ell))}} &\zeta_2\left[(U_{O_A}\otimes U_{O_B})|\Psi_O\rangle\right]\\ &= \prod_{i=1}^{\ell}\left(1-\lambda_i^2+\lambda_i^4\right)\ , \end{aligned} \tag{6}$$

the maximum being attained in the Botero–BCS canonical axes of Eq. (2).

The proof is given in full detail in SM, Sec. S1. It assumes that the state $|\Psi_O\rangle$ is pure and Gaussian, and uses that A and B are contiguous blocks, which guarantees that Eq. (4) amounts to a maximization of the stabilizer purity over the orbit of the covariance matrix under $O_A\oplus O_B$. The key point is that local Gaussian unitaries over a bipartition of the system bring any pure fermionic Gaussian state into a canonical modewise form $|\Psi_{\mathrm{can}}\rangle$ in Eq. (2), consisting of independent two-mode Bell-like pairs [40], a normal form that underlies several equivalent descriptions of fermionic Gaussian entanglement [56–58]. We prove that this representative is the *global* maximizer of the stabilizer purity ζ_2 on the local Gaussian orbit, i.e. $M_2^{\mathrm{GNL}}(|\Psi_O\rangle) = M_2(|\Psi_{\mathrm{can}}\rangle)$, from which Eq. (5) follows by direct computation of the two-mode stabilizer entropy [36, 37].

For integer $\alpha > 2$ we conjecture that the same structure survives, with the Rényi weight adapted accordingly,

$$\begin{aligned} M_\alpha^{\mathrm{GNL}}(|\Psi_O\rangle) &= \sum_{i=1}^{\ell}\mathfrak{m}_\alpha(\lambda_i^2)\ ,\\ \mathfrak{m}_\alpha(x) &:= \frac{1}{1-\alpha}\log_2\left[\frac{(1-x)^\alpha + 1 + x^\alpha}{2}\right]\ , \end{aligned} \tag{7}$$

which reduces to Eq. (5) at $\alpha = 2$. This conjecture is supported by the extensive variational optimization reported in the SM, Sec. S2, and it reproduces the same phenomenology as the $\alpha = 2$ case (SM, Sec. S4).

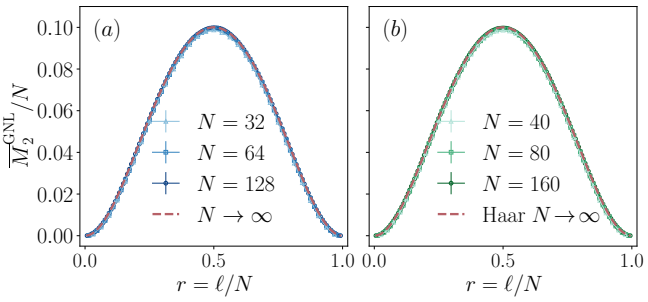


FIG. 1. **Gaussian-restricted non-local magic of typical fermionic Gaussian states.** (a) Page curve for Haar-random FGS: $\overline{M_2}^{\rm GNL}/N$ is plotted against the ratio $r = \ell/N$. The results are compared with the closed form expression for $N \to \infty$ in Eq. (8). (b) The Gaussian-restricted non-local magic Page curve is tested against the numerically computed average non-local magic for SYK_2 mid-spectrum eigenstates. The results are in quantitative agreement with Eq. (8).

Several remarks are in order. First, the proof of Theorem 1 relies heavily on the specific quadratic (collision) functional form. Proving the result for $\alpha > 2$ would require a different proof strategy, which we leave open for future investigations. Second, for $\ell \leq 2$, structural simplifications make the Gaussian minimization coincide with the full one, $M_2^{\rm NL} = M_2^{\rm GNL}$, and Eq. (5) reproduces the expression obtained in Ref. [36] from the Schmidt spectrum. For larger subsystems, non-Gaussian local transformations can further decrease the magic even when the initial state is Gaussian, so that Theorem 1 provides an upper bound on the generic non-local magic, $M_2^{\rm NL} \leq M_2^{\rm GNL}$. Third, $\mathfrak{m}_2(\lambda_j^2)$ vanishes both at $\lambda_j = 0$ (maximally entangled mode) and at $\lambda_j = 1$ (unentangled mode): entanglement is necessary but not sufficient for non-local magic, and a state can be maximally entangled yet carry zero nonstabilizerness across the bipartition, as is the case for the rainbow state (SM, Sec. S6).

Typical Gaussian states.—Having established the exact formula, we first investigate the Gaussian-restricted non-local magic of *typical* Gaussian states, i.e., states $|\Psi_O\rangle = U_O|\mathbf{0}\rangle$ generated by Haar-random $O \in \mathrm{O}(2N)$. Our goal is to compute the average $\overline{M_2}^{\rm GNL} \equiv \mathbb{E}_{O\sim \mathrm{O}(2N)}[M_2^{\rm GNL}(|\Psi_O\rangle)]$, which for our closed-form expression reduces to a random matrix theory problem [59–61]: the variables $x_k := \lambda_k^2$ form a Jacobi ensemble [59, 62], whose one-point density yields the exact average at finite size N (End Matter). We emphasize that this is a *quenched* average — performed outside the logarithm — which is typically harder to evaluate than its annealed counterpart.

In the thermodynamic limit $\ell, N \to \infty$ at fixed ratio $r = \ell/N \in [0, 1/2]$, the Gaussian-restricted non-local magic density $\overline{\mathfrak{m}}_2^{\rm GNL} := \overline{M_2}^{\rm GNL}/N$ (symmetric under $r \to 1-r$) follows from the free Jacobi law (End Matter),

$$\overline{\mathfrak{m}}_2^{\rm GNL}(r) \simeq \frac{c}{\pi}\int_0^{\pi/2} \frac{\cos^2\theta}{1 - c\sin^2\theta}\mathfrak{m}_2(c\sin^2\theta)d\theta\,, \qquad (8)$$

with $c = 4r(1-r)$. At exactly half bipartition, $r = 1/2$, this integral can be performed exactly and the formula reduces to $\overline{\mathfrak{m}}_2^{\rm GNL}(r) \simeq 2 - \log_2(2+\sqrt{3})$. On the other hand, for small $r \simeq 0$ (and by symmetry $r \simeq 1$), we have $c \simeq 4r$ and the integral yields $\overline{\mathfrak{m}}_2^{\rm GNL}(r \simeq 0) \simeq \frac{r^2}{\ln(2)} + \mathcal{O}(r^3)$. This *Gaussian-restricted non-local magic Page curve* shows a quadratic onset, for every α (SM, Sec. S4), with a transparent physical origin: when ℓ is of order one while $N \to \infty$, the ℓ modes of A are almost, or exactly, maximally entangled and thus carry no non-local magic. This is in sharp contrast with the entanglement Page curve, which vanishes only at $|A| = 0$ and grows linearly for small r [62–64].

To test the universality of this result beyond Haar-random states, we consider the Sachdev–Ye–Kitaev model at quadratic order, SYK_2, and its mid-spectrum eigenstates (SM, Sec. S5). In Fig. 1(b), we plot the resulting non-local magic density $\overline{\mathfrak{m}}_2^{\rm GNL}$ versus $r = \ell/N$ for $N = 40, 80, 160$. The data converge to the analytical Haar prediction Eq. (8) as N increases.

Ground states of the XY model.—We now apply Eq. (5) to study the equilibrium properties of Gaussian-restricted non-local magic in the ground state of the XY chain

$$H = -\sum_{j=1}^{N}\left[\frac{1+\eta}{2}X_jX_{j+1} + \frac{1-\eta}{2}Y_jY_{j+1} + \mu Z_j\right], \qquad (9)$$

with periodic boundary conditions. The XY chain is studied as a *fermionic* model, i.e. as a Majorana chain: after the Jordan–Wigner mapping (SM, Sec. S7) the ground state is naturally written — and exactly solvable — as a fermionic Gaussian state, and the natural notion of locality is that of the fermionic modes. Accordingly, $M_\alpha^{\rm GNL}$ is the non-local magic of the *free-fermion problem*, and the local Gaussian unitaries of Eq. (4) are, for a contiguous cut, implemented by genuine tensor products $U_{O_A} \otimes U_{O_B}$ of qubit unitaries. In the thermodynamic limit $N \to \infty$ with ℓ fixed, the reduced covariance matrix Γ_A of $A = \{1, \ldots, \ell\}$ is a $2\ell \times 2\ell$ block-Toeplitz matrix with 2×2 blocks depending only on $m-n$, whose symbol $g(k)$ is fixed by the Bogoliubov angle θ_k (SM, Sec. S7). The Gaussian-restricted non-local magic $M_2^{\rm GNL} = \sum_n \mathfrak{m}_2(\lambda_n^2)$ is therefore a spectral function of Γ_A with symbol $g(k)$, computable at classical cost $\mathcal{O}(\ell^3)$. Since the symbol satisfies $|\det g(k)| = 1$ for all k, the Szegő theorem rules out volume-law scaling of $M_2^{\rm GNL}$. When $g(k)$ is smooth, as occurs in the gapped phases $\mu \neq 1$ where θ_k is analytic, the strong Szegő theorem further implies that $M_2^{\rm GNL}$ saturates to a finite value as $\ell \to \infty$, with exponentially small corrections [65]. Logarithmic growth can arise only from singularities in the

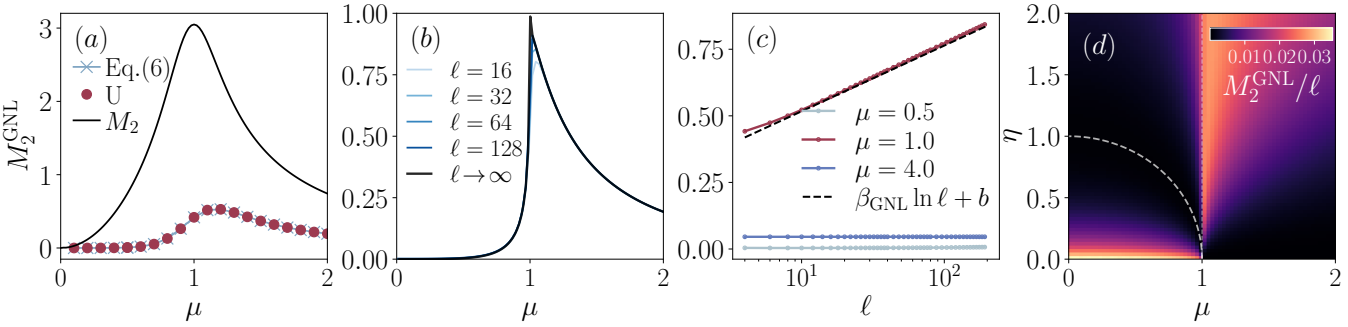


FIG. 2. **Equilibrium Gaussian-restricted non-local magic in the Ising and XY models.** (a) Validation of Eq. (5) for the Ising ground state with $N = 8$, $\ell = 4$: the Gaussian-restricted non-local magic M_2^{GNL} from Eq. (5) (crosses) agrees with the brute-force minimization over all local unitaries $U_A \otimes U_B$ (red marker) to numerical precision. The black line shows the total stabilizer entropy M_2 for reference. (b) M_2^{GNL} vs. transverse field μ in the thermodynamic limit for subsystem sizes $\ell = 16, 32, 64, 128$; the black curve is the exact $\ell \to \infty$ result, Eq. (11). (c) Scaling of $M_2^{\mathrm{GNL}}(\ell)$ at criticality ($\mu = 1$, red): the dashed line is the fit $\beta_{\mathrm{fit}} \ln \ell + b$ for $\ell > 50$, with $\beta_{\mathrm{fit}} \approx 0.1072$ and $b \approx 0.27$, in agreement with Eq. (12). In the gapped phases ($\mu = 0.5$, $\mu = 4.0$), M_2^{GNL} saturates to a finite value. (d) Heatmap of M_2^{GNL}/ℓ over the XY phase diagram (μ, η), computed from the exact Peschel formula.

symbol [66], which for the Ising model ($\eta = 1$) occur at the critical point $\mu = \mu_c = 1$, where the Bogoliubov angle θ_k develops a jump discontinuity at $k = 0$.

Away from criticality, the entanglement spectrum of a semi-infinite bipartition is known in closed form through the corner transfer matrix approach [67, 68]: the single-particle entanglement energies are equally spaced,

$$\varepsilon_n^{\mathrm{ent}} = \begin{cases} (2n+1)\,\varepsilon & \mu > 1\,, \\ 2n\,\varepsilon & \mu < 1\,, \end{cases} \tag{10}$$

where ε is the level spacing (SM, Sec. S7), and $\lambda_n = \tanh(\varepsilon_n^{\mathrm{ent}}/2)$ [69]. For a block of ℓ sites in the infinite chain with periodic boundary conditions, two separate boundaries contribute, so that

$$M_2^{\mathrm{GNL}}(\ell) \simeq 2 \sum_{n=0}^{\infty} \mathfrak{m}_2 \left[\tanh^2 \left(\frac{\varepsilon_n^{\mathrm{ent}}}{2} \right) \right], \tag{11}$$

up to corrections exponentially small in ℓ/ξ, where ξ is the correlation length of the problem.

In Fig. 2(a), we validate Eq. (5) by comparing against a brute-force minimization over the full unitary group $\mathrm{U}(d_A) \otimes \mathrm{U}(d_B)$ for $N = 8$, $\ell = 4$: the two results agree to numerical precision. For a general model (i.e., beyond Ising or XY), however, we do not expect such agreement, and only the inequality $M_2^{\mathrm{NL}} \leq M_2^{\mathrm{GNL}}$ is expected to hold, cf. SM, Sec. S2 [1]. In Fig. 2(b), we compare exact numerics for $N = 10^6$ qubits and $\ell = 16, 32, 64, 128$ with Eq. (11), finding quantitative agreement. Notably, for $\mu < 1$ the non-local magic is strongly suppressed (End Matter).

At the critical point $\mu_c = 1$ ($\eta = 1$), the gap closes and the spacing $\varepsilon \to 0$, so the series in Eq. (11) diverges. The key result is that the asymptotic density of the positive eigenvalues $\lambda_n \in [0, 1)$ of $i\Gamma_A$ — one per $\pm\lambda_n$ pair, with two Fisher–Hartwig boundary points for the block — scales as $\rho(\lambda) \simeq 2 \ln \ell \, [\pi^2(1-\lambda^2)]^{-1}$ [71]. Inserting this spectral density into Eq. (5) gives $M_2^{\mathrm{GNL}}(\ell) \simeq \beta_{\mathrm{GNL}} \ln \ell$ [Fig. 2(c)], with constant

$$\beta_{\mathrm{GNL}} = \frac{2}{\pi^2} \int_0^1 \frac{\mathfrak{m}_2(\lambda^2)}{1-\lambda^2}\, d\lambda = \frac{\pi^2 - \ln^2\left(7 - 4\sqrt{3}\right)}{\pi^2 \ln(16)}\,. \tag{12}$$

This logarithmic scaling is the non-local magic analogue of $S \simeq (c/3) \ln \ell$ [72], with the central charge c replaced by β_{GNL}. Over the full XY phase diagram [Fig. 2(d)], the Gaussian-restricted non-local magic is concentrated near the critical lines and is strongly suppressed deep in the gapped phases. These results are more general: Eq. (5) implies an area law for Gaussian-restricted non-local magic in ground states of one-dimensional local gapped free-fermion Hamiltonians (End Matter).

Quench dynamics and quasiparticle picture.— We now turn from equilibrium to the non-equilibrium dynamics of Gaussian-restricted non-local magic, focusing on quantum quenches in the Ising chain. The system is prepared in the ground state $|\Psi_0\rangle$ of $H(\mu_0)$ and, at time $t = 0$, the transverse field is suddenly quenched to $\mu \neq \mu_0$; the state then evolves unitarily as $|\Psi(t)\rangle = e^{-iH(\mu)t/2}|\Psi_0\rangle$, where, following Refs. [73, 74], we measure time in units in which the evolution is generated by $H(\mu)/2$. The state remains Gaussian, and $\Gamma_A(t)$ is block-Toeplitz with a time-dependent symbol [73] fixed by the post-quench dispersion ε_k and by the mismatch $\Delta\theta_k = \theta_k - \theta_k^{(0)}$ between the post-quench and pre-quench Bo-

[1] We nevertheless find that throughout the XY phase diagram, for the system sizes accessible to exact minimization, the agreement persists, see Ref. [70].

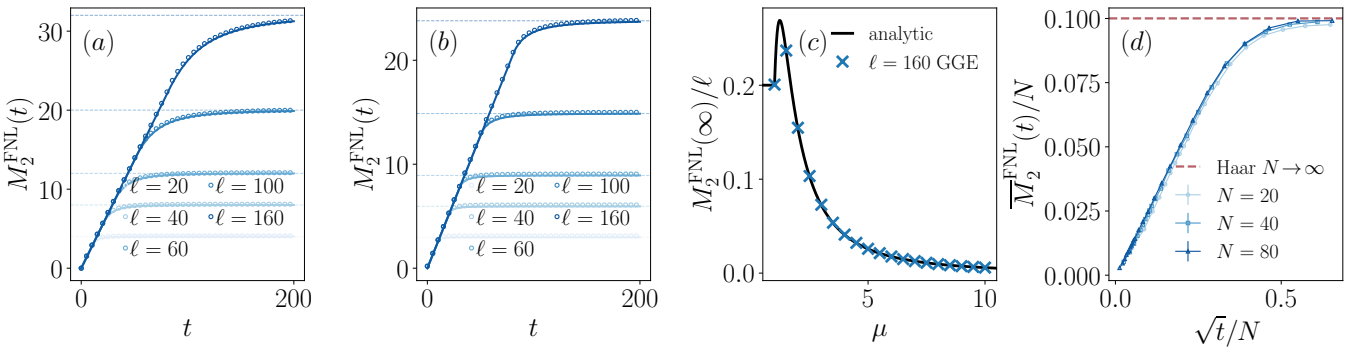


FIG. 3. **Dynamics of Gaussian-restricted non-local magic.** (a) Quench at $\eta = 1$ from the fully polarized state ($\mu_0 \to \infty$) to the critical point ($\mu = 1$): the exact numerics (circles) agrees quantitatively with the analytical prediction in Eq. (14) (solid lines) for subsystem sizes $\ell = 20, 40, 60, 100, 160$. Dashed lines mark the GGE stationary value. (b) Same for a finite quench $\mu_0 = 2 \to \mu = 1$. (c) Stationary-state density $M_2^{\mathrm{GNL}}(\infty)/\ell$ as a function of the post-quench field μ ($\mu_0 \to \infty$): analytical prediction from Eq. (13) (solid) and GGE numerics at $\ell = 160$ (crosses). (d) Random Gaussian circuit dynamics: $\overline{M_2}^{\mathrm{GNL}}/N$ is plotted against $\sqrt{t}/N$ for $N = 20, 40, 80$, exhibiting diffusive growth $\sim\sqrt{t}$ and saturation to the fermionic Gaussian Haar-random value $\overline{\mathfrak{m}}_2^{\mathrm{GNL}}(r = 1/2) = 2 - \log_2(2 + \sqrt{3})$ (dashed line).

goliubov angles (SM, Sec. S8). At long times $t \to \infty$, the oscillating terms of the symbol average to zero due to dephasing across modes with different group velocities [73]. The resulting stationary covariance matrix is that of a generalized Gibbs ensemble (GGE) [61, 73], whose eigenvalues are determined by the Bogoliubov angle mismatch alone, $\lambda_k^\infty = |\cos \Delta\theta_k|$, and, since $\mathfrak{m}_2(x) = \mathfrak{m}_2(1-x)$, the stationary Gaussian-restricted non-local magic is obtained via the Szegő theorem [73]

$$M_2^{\mathrm{GNL}}(\infty) = \frac{\ell}{\pi} \int_0^\pi dk\, \mathfrak{m}_2\big(\sin^2 \Delta\theta_k\big) \ . \tag{13}$$

This result reveals that for any $\mu \neq \mu_0$, the quench generically produces extensive (volume-law) Gaussian-restricted non-local magic; the expression is symmetric under $\mu \leftrightarrow \mu_0$.

This structure, closely parallel to the entanglement entropy of the same quench (End Matter), extends beyond the stationary state: writing M_2^{GNL} as a contour integral over the characteristic polynomial of $i\Gamma_A$ and evaluating the moments $\mathrm{Tr}(i\Gamma_A)^{2n}$ via a multidimensional stationary phase approximation [74] (SM, Sec. S8), we obtain the quasiparticle formula

$$M_2^{\mathrm{GNL}}(\ell, t) \simeq \int_0^\pi \frac{dk}{\pi}\, \min\big(2|v_k|\, t,\ \ell\big)\ \mathfrak{m}_2\big(\sin^2(\Delta\theta_k)\big), \tag{14}$$

valid in the scaling limit $\ell, t \gg 1$ with t/ℓ arbitrary, where $v_k = d\varepsilon_k/dk$ is the group velocity. Eq. (14) predicts linear growth for $t < t^\star = \ell/(2v_{\max})$, with $v_{\max} = \max_k |v_k|$, and saturation to Eq. (13) for $t \gg t^\star$. Eq. (14) has a transparent physical interpretation in terms of the quasiparticle picture [73, 74]: the quench creates pairs of entangled quasiparticles at each point, which propagate ballistically with mode-dependent velocities $\pm v_k$. A pair at momentum k contributes to M_2^{GNL} of the block A only when one partner has entered A while the other remains outside, hence the $\min(2|v_k|t, \ell)$ factor. The exact time evolution in Fig. 3(a,b) is in excellent agreement with the semiclassical prediction (solid lines) from Eq. (14), and the stationary values from Eq. (13) (dashed lines) are approached at $t \simeq t^\star$. In contrast, operator spreading in local random free-fermion circuits is *diffusive* [75, 76], with $\sqrt{t}$ growth of the Gaussian-restricted non-local magic [Fig. 3(d) and End Matter].

Summary.—We have proved an exact, closed-form expression for the fermionic Gaussian-restricted non-local magic of Gaussian states at Rényi index $\alpha = 2$. This establishes fermionic Gaussian states as a broad class of highly entangled many-body systems where the related non-local magic is exactly tractable, providing a useful proxy for the non-local magic over the full Unitary group. Beyond the physical result, the proof introduces two ingredients that we expect to be of independent interest: a sharp bound controlling the collision probability of a single Slater determinant by its transpose overlap; and a Hadamard-type determinant inequality for positive matrices of rank at most two, proved using fermionic creation operators. The analytical tractability established here opens several natural directions for future investigation (SM, Sec. S9), from higher-dimensional, disordered and topological free-fermion systems to symmetry-resolved, monitored and non-Hermitian settings.

Note added.—While preparing this manuscript, we discovered a closely related independent work by M. Collura, B. Beri, and E. Tirrito, appearing in the same arXiv posting [77]. When overlapping, our results are consistent.

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[55] See Supplemental Material at [URL will be inserted by publisher] for the full proof of Theorem 1, the variational verification of Eqs. (5) and (7), the error analysis for the experimental estimation, the generalization to Rényi index $\alpha > 2$, details on the numerical simulations, the rainbow-state example, details on the XY chain and on the quench dynamics, and an extended outlook, which include Refs. [78–83].

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END MATTER

Experimental observability.—A key practical advantage of Eq. (5) is that the Gaussian-restricted non-local magic of Gaussian states is entirely determined by the reduced covariance matrix Γ_A, a polynomial number of two-point correlation functions. While a proof-of-principle demonstration of non-local magic has recently been reported on a superconducting processor in a two-qubit setting [84], our result opens a scalable route applicable to large fermionic Gaussian systems.

A natural experimental protocol is *fermionic shadow tomography* [48]: random matchgate circuits are applied, followed by computational-basis measurements, and Γ_A is inferred from the resulting classical-shadow data. A conservative sample-complexity estimate yields, for failure probability at most δ,

$$N_{\rm shots} = O\left(\frac{\ell^2 \log(\ell/\delta)}{\epsilon_\Gamma^2}\right),$$

to reconstruct Γ_A with accuracy ϵ_Γ, while the propagation of this error to $M_2^{\rm GNL}$ is discussed in the SM, Sec. S3.

Finite-size Page curve.—For a fixed subsystem A of size $\ell \leq N/2$, let us define the stochastic variables $x_k := \lambda_k^2$, whose joint probability distribution is a Jacobi ensemble [59, 62]

$$\mathbb{P}(x_1,\dots,x_\ell) = \frac{\Delta(x_1,\dots,x_\ell)}{Z_\ell} \prod_{m=1}^{\ell} \frac{(1-x_m)^{N-2\ell}}{\sqrt{x_m}}\,, \quad (15)$$

with $\Delta(x_1,\dots,x_\ell) = \prod_{1\leq i<j\leq \ell}(x_i-x_j)^2$ the squared Vandermonde determinant and Z_ℓ the normalization ensuring $\int \mathbb{P}(x_1,\dots,x_\ell)\prod_{m=1}^{\ell}(dx_m) = 1$. Since $M_2^{\rm GNL} = \sum_{m=1}^{\ell} \mathfrak{m}_2(x_m)$ is a sum over separate x_m contributions, the only ingredient required is the one-point density function

$$\begin{aligned}\varrho^{(\ell)}(x) :&= \mathbb{E}_{\mathbb{P}}\left[\sum_{m=1}^{\ell}\delta(x-x_m)\right] \\ &= \ell \int \mathbb{P}(x,x_2,\dots,x_\ell)\prod_{m=2}^{\ell}(dx_m)\,,\end{aligned} \quad (16)$$

where we used relabelling in the second step. Notice that $\int_0^1 \varrho^{(\ell)}(x)dx = \ell$. Thus, the average Gaussian-restricted non-local magic is

$$\overline{M}_2^{\rm GNL} = \int_0^1 \mathfrak{m}_2(x)\varrho^{(\ell)}(x)dx\,. \quad (17)$$

For the Jacobi ensemble, the one-point density is given by the standard orthogonal-polynomial kernel. Given $w(x) = (1-x)^{N-2\ell}/\sqrt{x}$, we have [51, 52]

$$\varrho^{(\ell)}(x) = w(x)\sum_{n=0}^{\ell-1}[p_n(x)]^2\,, \quad (18)$$

where p_n are polynomials in the domain $[0,1]$, that are orthonormal with respect to the weight $w(x)$, namely

$p_n(x) = \mathrm{P}_n^{(-1/2,N-2\ell)}(1-2x)/\sqrt{h_n}$ in terms of the Jacobi polynomials $\mathrm{P}_n^{(a,b)}$ and with normalization

$$h_n = \frac{\Gamma(n+1/2)\Gamma(n+N-2\ell+1)}{(2n+N-2\ell+1/2)\Gamma(n+1)\Gamma(n+N-2\ell+1/2)}. \tag{19}$$

Eq. (18) yields the exact average non-local magic at finite size N for a given ℓ, which we benchmarked against direct sampling of Haar-random $O \in \mathrm{O}(2N)$, including the data of Fig. 1(a), finding agreement within statistical errors; the figure compares these data with the thermodynamic limit derived below. In the thermodynamic limit $\ell, N \to \infty$ at fixed ratio $r = \ell/N \in [0, 1/2]$, the rescaled one-point density $\varrho^{(\ell)}/N$ converges to the free Jacobi law with support $x \in [0, c]$, $c = 4r(1-r)$,

$$\varrho_r(x) = \frac{\sqrt{(c-x)x}}{2\pi x(1-x)} \mathbf{1}_{[0,c]}(x)\,, \tag{20}$$

where $\mathbf{1}_A(x)$ is the characteristic function of A, $\mathbf{1}_A(x) = 1$ if and only if $x \in A$. The total mass of the density is $\int_0^c \varrho_r(x)dx = r$, which is precisely the fraction of the number of eigenvalues over the system size. Thus, in the scaling limit, we have

$$\overline{\mathfrak{m}}_2^{\mathrm{GNL}}(r) \simeq \int_0^c \mathfrak{m}_2(x)\varrho_r(x)dx\,. \tag{21}$$

A more convenient parametrization is obtained by setting $x = c\sin^2\theta$ with $\theta \in [0, \pi/2]$, which gives Eq. (8).

Ordered phase of the Ising chain.—For $\mu < 1$ the non-local magic is strongly suppressed, M_2^{GNL} being exponentially small in the level spacing ε of the entanglement spectrum and vanishing exactly only in the GHZ limit $\mu \to 0$. This is the ordered (ferromagnetic) phase, whose ground state approaches the GHZ state $\frac{1}{\sqrt{2}}(|+\rangle^{\otimes N} + |-\rangle^{\otimes N})$, with $X|\pm\rangle = \pm|\pm\rangle$, as $\mu \to 0$. The suppression of non-local magic in this regime can be understood from the Peschel entanglement spectrum, Eq. (10): for $\mu < 1$ and $\eta = 1$, the single-particle entanglement energies are $\varepsilon_n^{\mathrm{ent}} = 2n\varepsilon$. The $n = 0$ mode has $\varepsilon_0^{\mathrm{ent}} = 0$, yielding $\lambda_0 = \tanh(0) = 0$ (a maximally entangled, stabilizer-like Bell pair), while all $n \geq 1$ modes have $\lambda_n = \tanh(n\varepsilon)$ exponentially close to 1 (nearly unentangled modes). Since $\mathfrak{m}_2(0) = \mathfrak{m}_2(1) = 0$, every mode contributes an exponentially small amount of non-local magic. This is a striking example of a phase with nonzero entanglement entropy (area law) but negligible non-local magic. In contrast, the entanglement entropy receives a finite area-law contribution from the $n = 0$ maximally entangled mode.

Area law for Gaussian-restricted non-local magic.—We present the argument for arbitrary $\alpha \geq 1$; it is rigorous at $\alpha = 2$ by Theorem 1, while for $\alpha \neq 2$ it inherits the status of the conjectured weights of Eq. (7). Let $|\Psi_O\rangle$ be the ground state of a one-dimensional local gapped free-fermion Hamiltonian and let $A = \{1, \ldots, \ell\}$ be a contiguous block, as in Theorem 1. The Gaussian entanglement entropy is

$$S(A) = \sum_{i=1}^{\ell} \mathfrak{f}\left(\frac{1+\lambda_i}{2}\right), \tag{22}$$

where $\mathfrak{f}(x) := -x\log_2 x - (1-x)\log_2(1-x)$. On the other hand, taking the limit $\alpha \to 1$ in Eq. (7) gives

$$M_1^{\mathrm{GNL}}(|\Psi_O\rangle) = \frac{1}{2}\sum_{i=1}^{\ell} \mathfrak{f}(\lambda_i^2). \tag{23}$$

Moreover, $M_\alpha^{\mathrm{GNL}} \leq M_\beta^{\mathrm{GNL}}$ for $\alpha \geq \beta$, since Rényi entropies are non-increasing in the index. By noticing that for every $\lambda \in [0, 1)$

$$\frac{1}{2}\mathfrak{f}(\lambda^2) \leq 2\,\mathfrak{f}\left(\frac{1+\lambda}{2}\right), \tag{24}$$

then, summing over the single-particle entanglement modes gives

$$M_\alpha^{\mathrm{GNL}}(|\Psi_O\rangle) \leq M_1^{\mathrm{GNL}}(|\Psi_O\rangle) \leq 2\,S(A), \qquad \alpha \geq 1. \tag{25}$$

This is the Gaussian analogue of the general entanglement-controlled bound discussed for M_2^{NL} in Ref. [36]. Finally, by Hastings' theorem [85], the entanglement entropy of a unique gapped ground state of a one-dimensional local Hamiltonian obeys

$$S(A) \leq C_{\mathrm{ent}}|\partial A|, \tag{26}$$

with C_{ent} independent of $|A|$ and of the total system size. Therefore

$$M_\alpha^{\mathrm{NL}}(|\Psi_O\rangle) \leq M_\alpha^{\mathrm{GNL}}(|\Psi_O\rangle) \leq 2\,C_{\mathrm{ent}}|\partial A|, \tag{27}$$

whenever $\alpha \geq 1$.

Quench dynamics: analogy with the entanglement entropy.—Eq. (13) bears a striking structural similarity to the entanglement entropy of the same quench [73], $S = \frac{\ell}{\pi}\int_0^\pi dk\ \mathfrak{f}\big(\sin^2(\Delta\theta_k/2)\big)$, with $\mathfrak{f}$ the binary entropy: in terms of the stationary eigenvalues $\lambda_k^\infty = |\cos\Delta\theta_k|$, the entanglement weight $\mathfrak{f}\big((1+\lambda_k^\infty)/2\big)$ is replaced by the magic weight $\mathfrak{m}_2\big((\lambda_k^\infty)^2\big) = \mathfrak{m}_2(\sin^2\Delta\theta_k)$, using the symmetry $\mathfrak{m}_2(x) = \mathfrak{m}_2(1-x)$. This parallel extends beyond the stationary state and allows us to derive, from first principles, a quasiparticle picture for the spreading of Gaussian-restricted non-local magic.

Random Gaussian circuit dynamics.—We consider the dynamics of Gaussian-restricted non-local magic under random Gaussian circuits [86]. The circuit is composed of nearest-neighbor fermionic Gaussian unitaries arranged in a brick-wall geometry on N sites. A single layer $O_{\mathrm{layer}} = O_{\mathrm{odd}}\,O_{\mathrm{even}}$ consists of an even and an odd half-step, namely $O_{\mathrm{even}} = \prod_{j=1}^{N/2} R_{2j-1,2j}$ and

$O_{\rm odd} = \prod_{j=1}^{N/2-1} R_{2j,2j+1}$, where each $R_{j,j+1} \in \mathrm{SO}(4)$ is a random orthogonal matrix drawn independently from the Haar measure, acting on the four Majorana operators $(\gamma_{2j-1}, \gamma_{2j}, \gamma_{2j+1}, \gamma_{2j+2})$ of sites j and $j+1$. The full circuit after t layers is $O(t) = \prod_{s=1}^{t} O_{\rm layer}^{(s)} \in \mathrm{SO}(2N)$. The initial state is the product state $|\mathbf{0}\rangle$ (vacuum of all fermionic modes), with covariance matrix Γ_0. The time-evolved state remains Gaussian at all times, with covariance matrix $\Gamma(t) = O(t)\,\Gamma_0\,O(t)^T$. In contrast to the ballistic quasiparticle spreading found in the integrable quench setting, operator spreading in local random free-fermion circuits is *diffusive* [75, 76]. Since the Gaussian-restricted non-local magic is a smooth spectral functional of $\Gamma_A(t)$, it inherits this diffusive transport. At early times the modes of A are nearly unentangled; expanding the single-mode contribution for $\lambda \to 1$, $\mathfrak{m}_2(\lambda^2) \simeq \frac{1-\lambda^2}{\ln 2}$, so that $M_2^{\rm GNL}(t) \simeq \frac{1}{2\ln 2}\{2\ell - \mathrm{Tr}[(i\Gamma_A(t))^2]\} = \frac{1}{2\ln 2}\|\Gamma_{AB}(t)\|_{\rm HS}^2$, with Γ_{AB} the off-diagonal block of Γ, which is governed by the diffusive spreading of two-point correlations across the entanglement cut [75]. This argument yields

$$M_2^{\rm GNL}(t) \propto \sqrt{t}\,, \qquad 1 \ll t \ll N^2\,. \tag{28}$$

At late times, $M_2^{\rm GNL}$ saturates to the Haar-random fermionic Gaussian value, and the natural scaling variable is $\sqrt{t}/N$. In Fig. 3(d), we plot $\overline{M_2}^{\rm GNL}(t)/N$ versus $\sqrt{t}/N$ for $N = 20, 40, 80$ and the half-chain bipartition, finding an excellent data collapse onto a single scaling function that interpolates between $\sqrt{t}$ growth and saturation at the Page-curve value $\overline{\mathsf{m}}_2^{\rm GNL}(r = 1/2) = 2 - \log_2(2+\sqrt{3})$ (dashed line). The saturation timescale $t \sim \mathcal{O}(N^2)$ is consistent with diffusive dynamics.